

```
int sum(int A[], int n)
```

```
{
    int sum = 0, n; → ①
```

```
    for(i = 0; i < n; i++) → ②
```

```
        sum = sum + A[i] → ③
```

```
    return sum; → ④
```

$2 + 4n$
 ~~$2 + 5n$~~

Cost	Repetition	Total
1	1	1
1 + 1 + 1	1 + (n + 1) + n	2n + 2
2	n	2n
1	1	1

Home Work

Compute the time complexity using frequency count for following snippets

```
for(i = 1; i <= 10; i = i + 2).
```

```
{
    i = i + 10;
    printf("%d", i);
}
```

```
i = 1
while(i <= 50)
{
    printf("hello");
}
```


for(i=1; i<=20; i++);

for(i=1; i<=n; i=i++);

```
{  
    printf("hello");  
    printf("world");  
    printf("i=%d", i);  
}
```

1.)

$$1+6+5+5+5 \Rightarrow \cancel{32} 32.$$

2.) ~~1~~ 1

$$3.) \cancel{1+2+20} \Rightarrow \cancel{42} \quad 1+1+1=3$$

$$4.) 1+(n+1)+n+3n = 5n+2$$